National Aeronautics and
Space Administration

Synthetic transcriptomics for mouse spaceflight

Can generated RNA-seq improve FLT vs GC analysis within each tissue?

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Biological & Physical Sciences

QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

1,610 biological profiles

75 accessions

835 FLT

775 GC

ARCHS4 mouse

997,515 profiles audited

17,244 selected

20 tissues

- Mission, strain, material and assay differences can resemble a flight effect.
- The generator must preserve tissue and FLT/GC structure without memorizing samples.
- Statistical evidence must still come from observed OSDR profiles.

Sources: NASA OSDR Biological Data API; ARCHS4 mouse v2.5.

METHOD

We built a configurable bulk RNA-seq generation pipeline

Each stage exposes alternatives; a strong outline marks the configuration used downstream.

1 Data scope

Sources
- **OSDR API**
- **ARCHS4**

Studies
- Single
- **Multiple**

Tissues
- Per tissue
- **All tissues**

2 Transform

Expression
- Raw
- CPM
- **TPM**

Scaling
- None
- Z-score
- **MaxAbs**

Features
- All
- HVG
- **L1000**

3 Harmonization

Global or study correction
- **None**
- Within-study z-score
- ComBat / MBatch
- MOBER

4 Model training

Generator
- WGAN-GP
- **DDIM**

Training source
- OSDR only
- ARCHS4 only
- **Pretrain + adapt**

5 Conditioning

Model inputs
- **Tissue**
- **FLT / GC**
- **Accession**
- **Material type**
- Sex / age

Selected branch

No global correction

ARCHS4 pretrain → **OSDR adaptation** → **Conditional DDIM**

TPM | MaxAbs | 974 landmarks

Tissue | FLT / GC | Accession | Material type

Screen: 463 planned configurations and nine matched liver harmonization arms. Models: Vinas et al. (2022); Lacan et al. (2026).

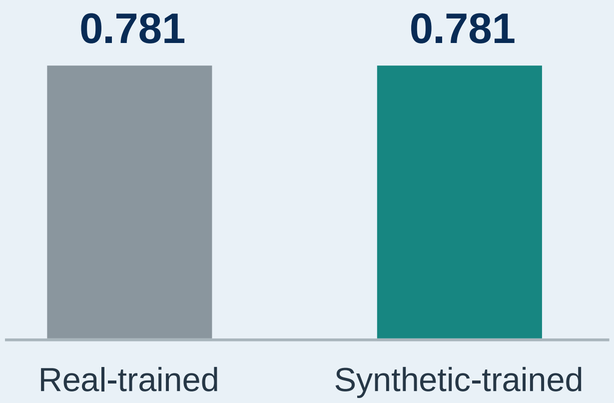
VALIDATION

DDIM matched expression and reduced separability

AA closer to 0.5 means a classifier has less success separating real and generated profiles; lower FD is better.

ARCHS4 tissue probe

Balanced accuracy on held-out real profiles



Metric	WGAN-GP	DDIM	Reading
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial accuracy	0.636	0.475	closer to 0.5
FD / real P95	0.144	0.074	lower

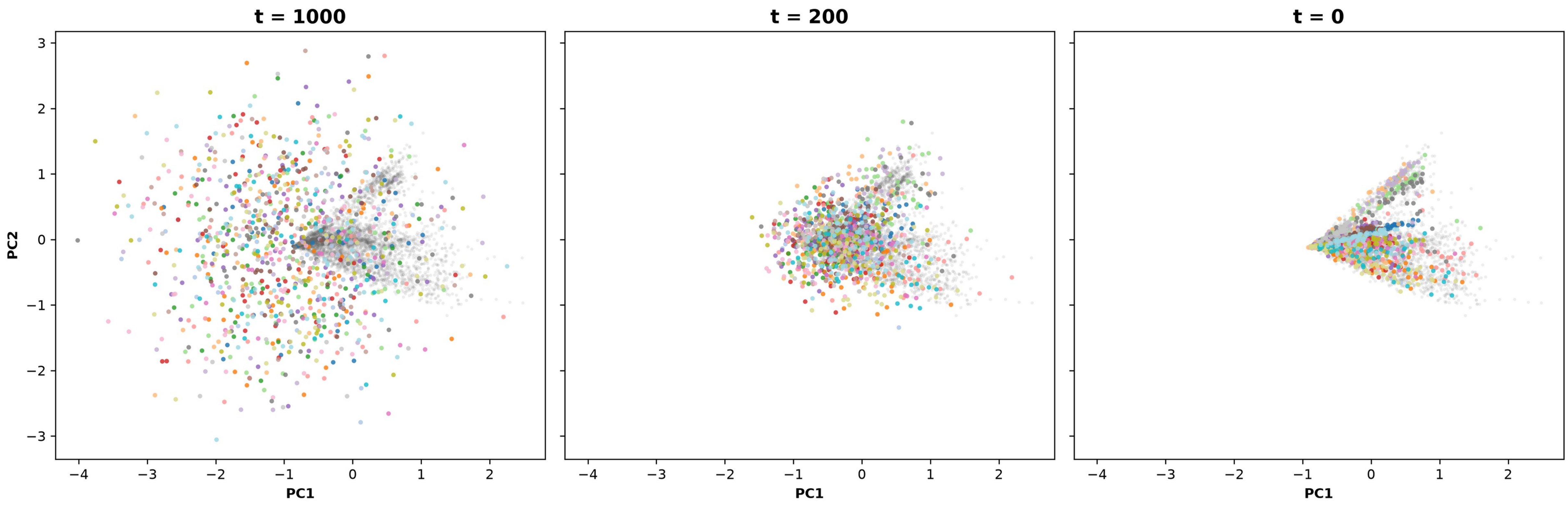
Use DDIM downstream

Lower separability and distributional distance, with comparable correlation and higher F1.

WGAN values use validation data and DDIM values use the stated OSDR test; they are not paired on one common split.

Diffusion learns tissue structure from noise

The same PCA axes follow 1,024 generated profiles through reverse diffusion.



t = 1000: noise

t = 200: partial structure

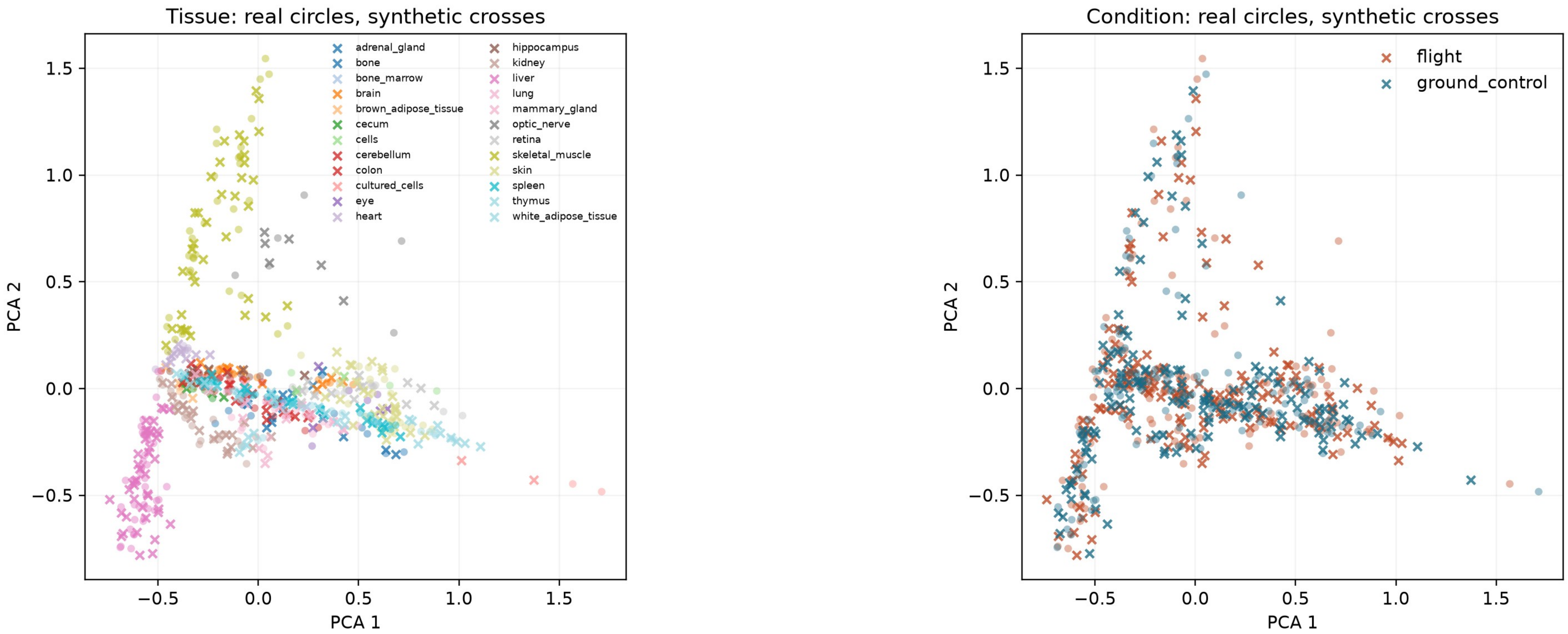
t = 0: tissue-conditioned profiles

Gray points are real ARCHS4 profiles; colors identify generated tissue conditions. PC1 and PC2 limits are shared across panels.

DIFFUSION OUTPUT

Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.



Interpretation: tissue structure dominates; FLT and GC are subtler. PCA is descriptive, so quantitative metrics determine validation.

PCA was fitted in the common locked OSDR expression space. This is PCA, not UMAP.

ANALYSIS

Synthetic profiles entered the analysis in five different ways

Every arm was judged on held-out real profiles before synthetic attribution was allowed.

Real OSDR
observed FLT / GC

DDIM generated
matched conditions

Real only
rank: real
fit: real

Generated only
rank: generated
fit: generated

Real + generated
rank: consensus
fit: equal weight

Guided, real fit
rank: consensus
fit: real

Guided, 5% synth.
rank: consensus
fit: real + 5%

Held-out real profiles

BA | AUROC | AP

select one eligible arm
for each tissue

Biology check FLT vs GC effects, random-effects meta-analysis and BH FDR use real OSDR profiles only.

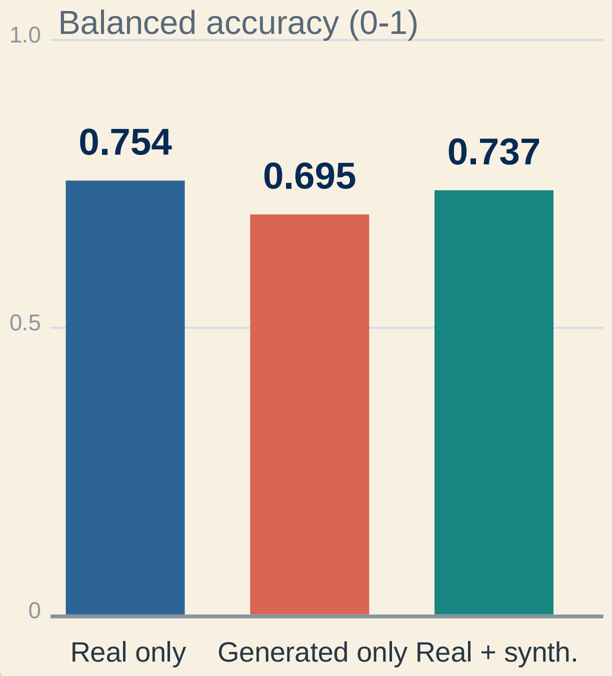
Synthetic profiles never increase animal n.

UTILITY

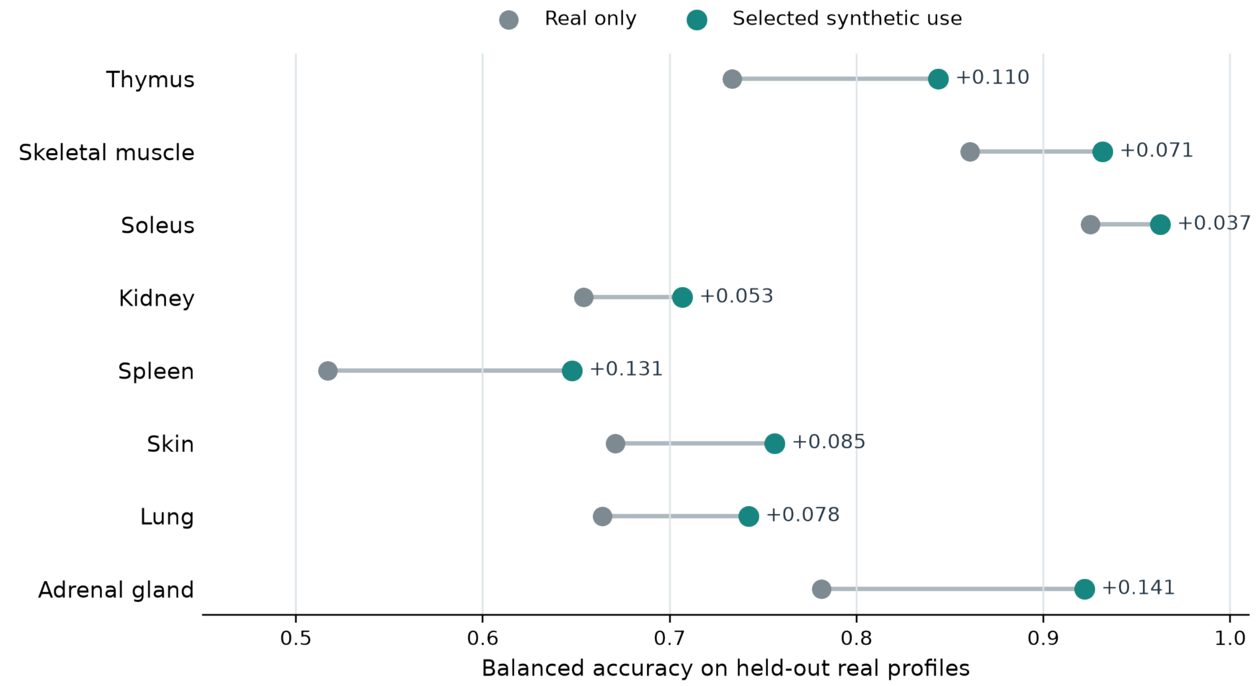
Pooling tissues hid useful signal

A single FLT/GC classifier did not improve, but tissue-specific synthetic use often did.

Pooled across tissues



Selected use within each tissue



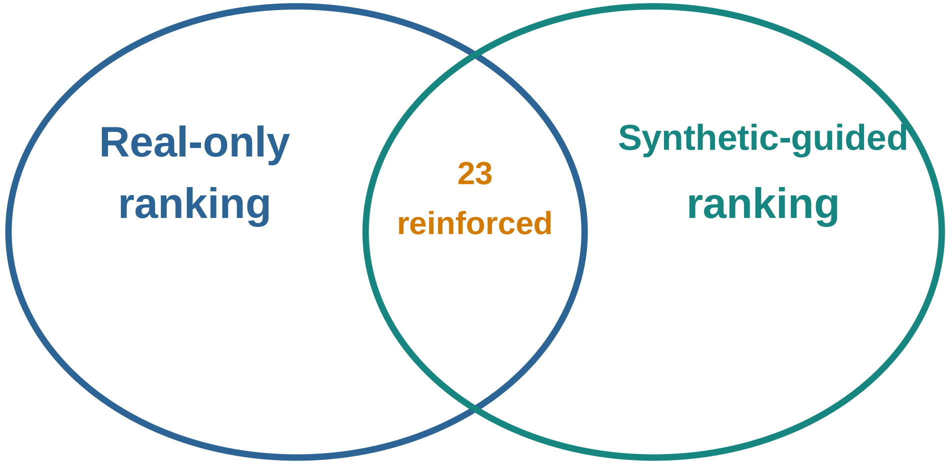
The useful arm differed by tissue.

Development results with held-out profiles from represented accessions; they do not measure transfer to a new mission.

Synthetic guidance changed ranking, not statistical evidence

Promoted and reinforced describe repeated feature selection; both require a supporting effect in real OSDR data.

Repeated stable selection



26 promoted
selected only with synthetic guidance

Real OSDR association test

- Estimate FLT vs GC inside each accession
- Combine accession effects with a random-effects model
- Apply BH FDR within each tissue

49 synthetic-informed associations

These are 49 tissue-gene associations among 459 BH-FDR rows. A gene can appear in more than one tissue.

LITERATURE INTERPRETATION

Selection and literature are separate dimensions

All 49 synthetic-informed associations received both a selection status and a literature interpretation.

1 Candidate
Gene + tissue +
FLT direction

2 Search
Spaceflight +
mechanism literature

3 Classify
Relation to
published evidence

4 Interpret
Recovery, extension,
or unmatched

Selection status

26 Promoted
Stable only with synthetic guidance

23 Reinforced
Stable with and without guidance

Literature interpretation

22	Aligning	Direct or same-tissue process agreement
19	Complementary	Related process or mechanism
4	Ambiguous	Mixed or context-dependent evidence
4	Unmatched	No sufficiently specific match found

A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.

Targeted review completed 2026-08-03; 5 direct transcript-level matches and a 33-source inventory are in Tables S16-S17.

ANALYSIS COVERAGE

All 27 completed tissue analyses were retained

The biological narrative narrows only after reporting synthetic-informed, real-only, and null outcomes.

10 Synthetic-informed

Promoted or reinforced BH-FDR gene

- Adrenal Gland
- Eye
- Kidney
- Skeletal muscle (pooled)
- Skin
- Spleen
- Thymus
- Gastrocnemius
- Soleus
- Tibialis Anterior

5 Real BH FDR only

Real association, no synthetic-informed selection

- Heart
- Liver
- Retina
- EDL
- Quadriceps

12 No panel BH-FDR gene

No BH-FDR gene in the 974-gene panel

- Bone
- Bone Marrow
- Brain
- Brown Adipose Tissue
- Cecum
- Cerebellum
- Colon
- Hippocampus
- Lung
- Mammary Gland
- Optic Nerve
- White Adipose Tissue

Twenty-two canonical tissues plus EDL, gastrocnemius, quadriceps, soleus and tibialis anterior; complete counts in Table S12.

GENE INVENTORY

Synthetic-informed genes spanned 10 tissue analyses

All 49 associations passed BH FDR in real OSDR data; synthetic profiles affected prioritization, not the statistical test.

[P] promoted [R] reinforced — aligning — complementary — ambiguous — unmatched *Tag = selection | Gene color = literature*

Thymus
16 genes
- FLT higher: [P]*Hsd17b11*, [P]*Etv1*, [R]*Snx7*
- FLT lower: [P]*Nusap1*, [P]*Stmn1*, [P]*Birc5*, [P]*Cdk1*, [P]*Top2a*, [P]*Ccnb2*, [P]*Aurka*, [P]*Ccne2*, [P]*Kif20a*, [P]*Pcna*, [P]*Ccnf*, [R]*Ube2c*, [R]*Gmnn*

Spleen
4 genes
- FLT higher: [P]*Rai14*, [P]*Myl9*, [P]*Ptprk*, [R]*Loxl1*
- FLT lower: *none*

Kidney
2 genes
- FLT higher: [P]*Inpp4b*, [R]*Slc37a4*
- FLT lower: *none*

Gastrocnemius
2 genes
- FLT higher: [P]*Nfkbia*
- FLT lower: [P]*Fhl2*

Eye
1 gene
- FLT higher: *none*
- FLT lower: [R]*Klhl21*

Skeletal muscle (pooled)
12 genes
- FLT higher: [R]*Sox4*, [R]*Cebpd*, [R]*Sh3bp5*, [R]*Prkcd*, [R]*Arid5b*, [R]*Sesn1*, [R]*Tle1*
- FLT lower: [P]*Klhl21*, [P]*Mapkapk5*, [P]*Reep5*, [P]*Itgb5*, [R]*Bphl*

Soleus
5 genes
- FLT higher: [R]*Tpm1*
- FLT lower: [R]*Bdh1*, [R]*Ech1*, [R]*Bnip3*, [R]*Decr1*

Tibialis anterior
4 genes
- FLT higher: [P]*Cebpd*, [R]*Cdkn1a*, [R]*St3gal5*, [R]*Bnip3*
- FLT lower: *none*

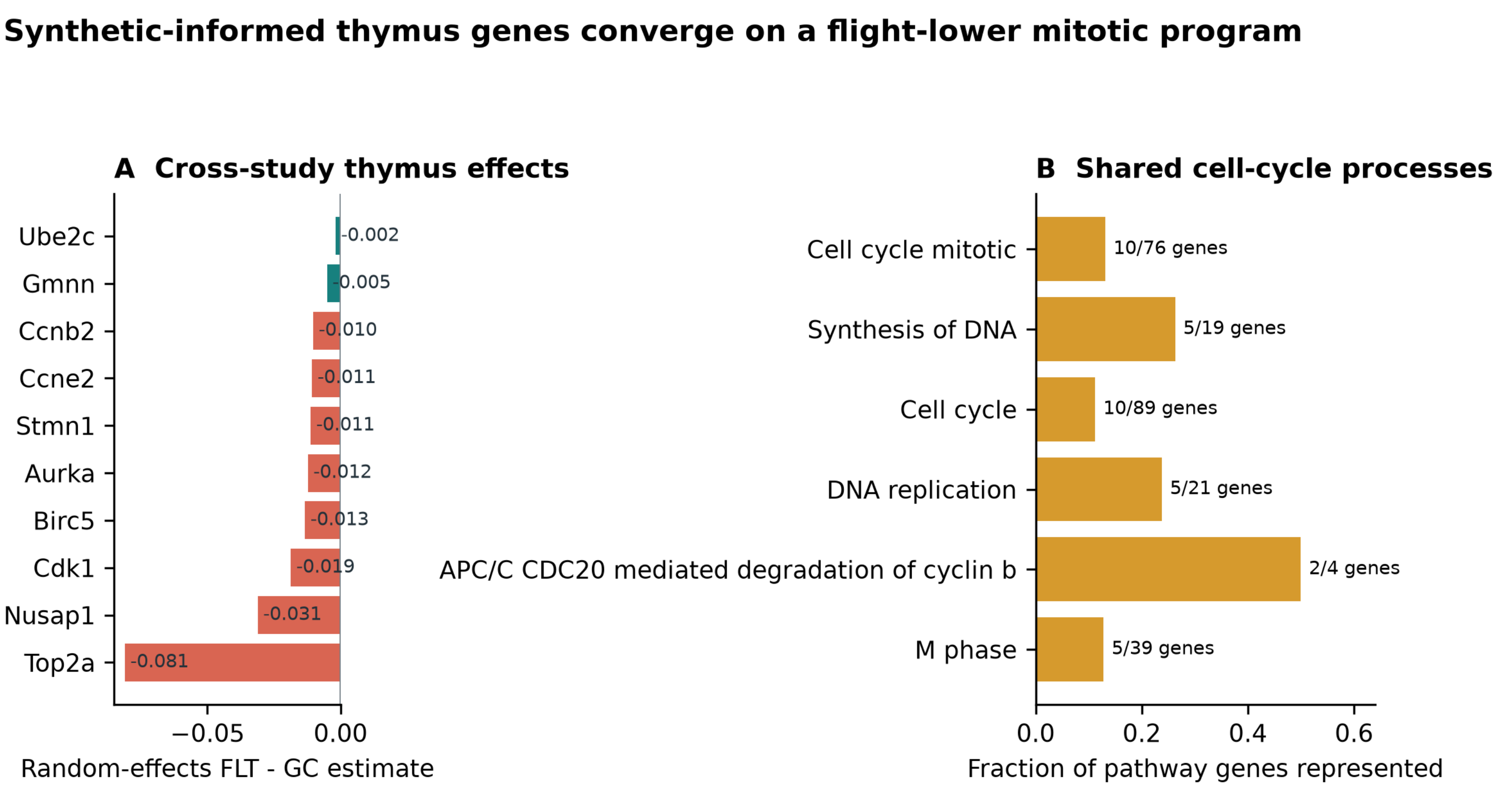
Adrenal gland
2 genes
- FLT higher: *none*
- FLT lower: [P]*Psmb8*, [R]*Tspan4*

Skin
1 gene
- FLT higher: [P]*Plscr1*
- FLT lower: *none*

[P]/[R] tags report repeated feature selection; gene color independently reports literature interpretation for all 49 associations.

Thymus points to lower proliferative renewal

The core panel recovers cell-cycle biology; two flight-higher genes extend the hypothesis.



16 synthetic-informed BH-FDR associations

13 promoted | 3 reinforced

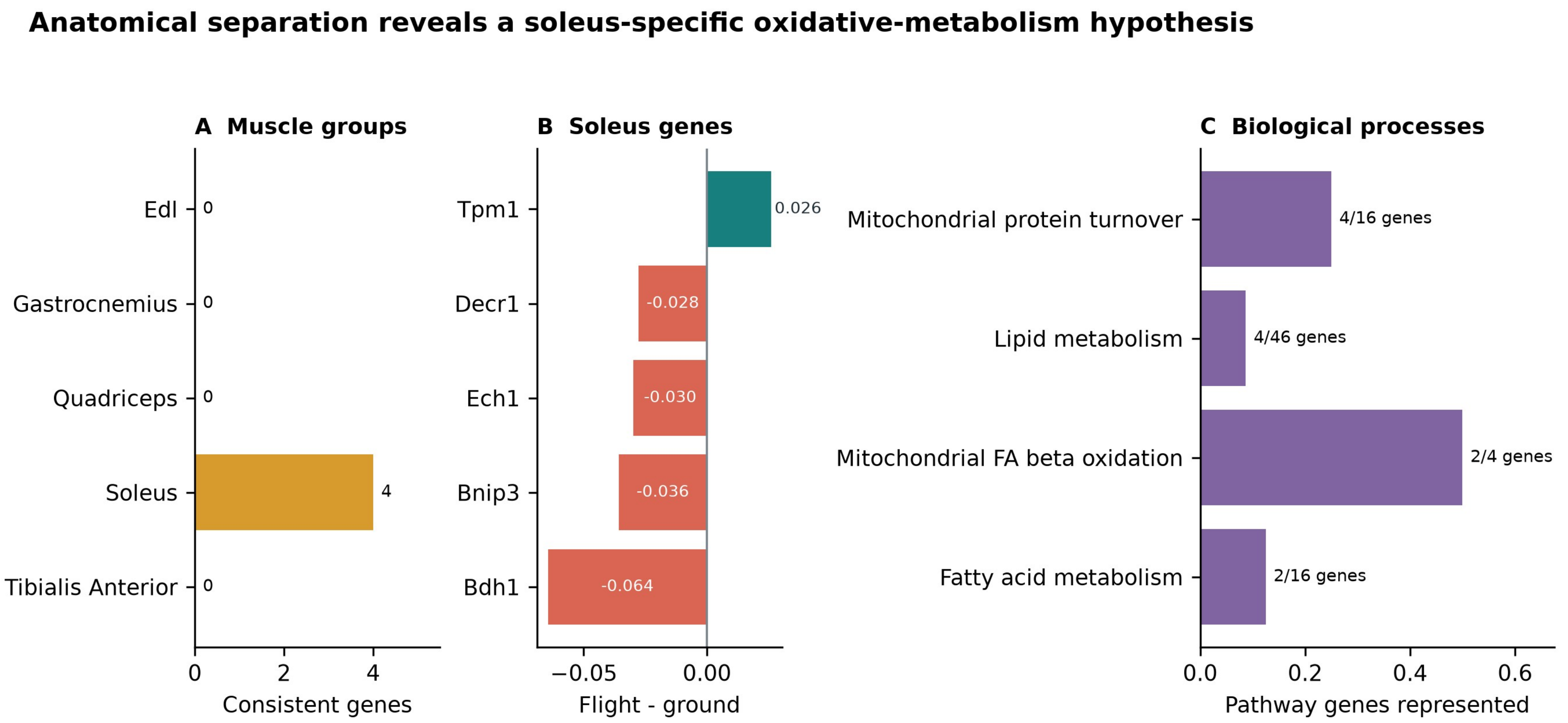
- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

Context: Gridley et al. (2013), Horie et al. (2019), Keenan et al. (2025), and Shi et al. (2026).

SOLEUS

Soleus reinforces a mitochondrial and lipid program

This result strengthens an existing real-data pattern rather than promoting a new soleus gene.



0.925 -> 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

Literature context: Gambara et al. (2017) and Stein et al. (2002).

ADDITIONAL FINDINGS I

Each additional tissue defines a separate hypothesis

Promoted and reinforced selections are separated within every tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Pooled muscle	Promoted	Lower: Klhl21, Mapkapk5, Reep5, Itgb5	Heterogeneous stress, differentiation, interferon and sialic-acid response; interpret with the anatomical muscle groups.
	Reinforced	Higher: Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1; lower: Bphl	
Kidney	Promoted	Inpp4b higher	Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.
	Reinforced	Slc37a4 higher	
Spleen	Promoted	Rai14, Myl9, Ptprk higher	Adhesion, actomyosin and extracellular-matrix or immune-organization hypothesis; no coherent pathway enrichment.
	Reinforced	Loxl1 higher	
Skin	Promoted	Plscr1 higher	Interferon-linked skin candidate within broader cell-cycle and DNA-repair responses; a single-gene result.
	Reinforced	None	

All listed effects passed BH FDR in observed OSDR profiles; 'None' means no gene in that selection category.

ADDITIONAL FINDINGS II

Eye, adrenal and muscle-group results remain tissue-specific

Promoted and reinforced selections are separated within every tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Eye	Promoted	None	Process-level alignment with lower proliferation and cytokinesis in flight eye tissue; not an exact prior gene replication.
	Reinforced	Klhl21 lower	
Adrenal gland	Promoted	Psmb8 lower	Literature-unmatched immune, proteostasis or tissue-composition candidates; no adrenal mechanism is established.
	Reinforced	Tspan4 lower	
Gastrocnemius	Promoted	Nfkbia higher; Fhl2 lower	NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.
	Reinforced	None	
Tibialis anterior	Promoted	Cebpd higher	Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.
	Reinforced	Cdkn1a, St3gal5, Bnip3 higher	

Interpretations follow the independent literature annotations in Table S16; 'None' means no gene in that selection category.

TAKEAWAYS

Synthetic data helped most as a tissue-specific prior

The generator created useful structure, not new biological replication.

1 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

2 Use

Pooled augmentation failed. Tissue-specific ranking and low-weight training were more useful.

3 Interpret

Thymus and soleus were clearest. Literature review separated recovery from complementary hypotheses.

Next test

Use independent samples and cell-resolved assays to test thymus proliferation, soleus metabolism and prioritized candidates from additional tissues.

Generated profiles never enter the biological sample count or the BH-FDR test.

Thank you

Questions?

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All biological associations were tested in observed OSDR profiles.